



DO-IT-YOURSELF: PROJECT

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2. Once the face appears sharp and focused on the screen, press Learn at the bottom-left of the TFT screen.

3. The face is saved in the system. (This can be verified by checking the increased file size of ``/root/faces.bin``.)

Step 2: Real-time operation

Once enrolled, the system functions as follows:

1. Employees stand in front of the camera.

2. If the system recognises the face from the database:

- The green LED lights up, indicating 'Gate Open.'

- The gate mechanism activates, allowing entry.

3. If the system does not recognise the face:

- The red LED lights up, indicating 'Gate Closed.'

Data management. The system maintains a *person.txt* file that maps each employee's real name to their corresponding ID in the database.

The employee's name and ID number are displayed on the screen during real-time operation.

Modifying or deleting entries. To modify or delete employee data the provided Python script can be used to edit entries in headless mode (SSH or serial connection). Alternatively, the *person.txt* file can be manually edited to update names.

Key features summary.

Simple enrollment: Register faces by pressing Learn.

Automated Gate operation: Facial recognition controls access.

Visual indicators: Red/green LEDs provide clear status feedback.

Editable database: Employee records can be managed with built-in tools.

Aftermath

With advancements in AI software (for example, YOLOv5→YOLOv8), hardware capabilities are improving to handle more complex tasks.

A logical next step for this device is to develop a log register that records legitimate entries with date and timestamp. This feature can be implemented as an enhancement. **EFY**

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